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Title: Unpacking the outcomes and trade-offs of Integrated Soil Fertility Management with probabilistic modeling approaches

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Abstract:

Sustainable intensification practices aim to increase agricultural productivity and farmer profits on existing land while minimizing negative environmental and social impacts. However, selecting the best practices to scale requires robust decision-making due to the complexity of agricultural systems and the risk of unintended consequences for vulnerable groups. Integrated Soil Fertility Management (ISFM) is being promoted in Northern Ghana as a sustainable intensification practice. However, ISFM outcomes and trade-offs across the dimensions of the Sustainable Intensification Assessment Framework (SIAF) (productivity, economic, social, and environmental domains) remain unclear. Because current assessment approaches of the outcomes of agricultural practices often fail to capture the complexity of agricultural systems in a holistic manner, we propose a probabilistic modeling approach to assess ISFM. This modelling approach incorporates risks, uncertainties, costs, and benefits of ISFM and uses Monte Carlo simulations to simulate and forecast the SIAF outcomes of ISFM in maize and soybean systems. By capturing the complexity of agricultural systems and providing evidence-based insights, our findings inform decisions on ISFM adoption and policy interventions, ensuring that trade-offs are acknowledged and equitable agricultural practices are promoted.

Keywords: Decision analysis, impact-oriented research, sustainable land management